Chapter 1

Problem, Context and Requirements

1.1 The problem

Floods are Tunisia's deadliest natural hazard: between 1957 and 2018 they caused about 70% of disaster casualties and 60% of economic losses [1], and they cost the country about US$40 million a year [2]. Nabeul, in Cap Bon, is a clear example. On 22 September 2018 about 200 mm of rain fell on Nabeul and up to 225 mm at Beni Khaled, the most since INM records began in 1995; six people died and more than 6,000 families were affected [3], with recovery needs of about US$100 million [1]. In January 2026 some regions had their heaviest rain since 1950, AFP reported five deaths and 350 people rescued, and Nabeul was again among the hardest-hit regions [4].

The warning existed; acting on it failed. A warning was issued on 21 September 2018, the day before the flood [3]. People still died, and some residents stayed on rooftops rather than cross flooded areas to reach evacuation points [3]. The weak step was not the forecast but the last mile: getting a clear, local instruction to the right people and making sure someone acted on it. The World Bank ranks Tunisia 13th of 19 MENA countries for early-warning services and names *inefficient procedures for alert dissemination* among the gaps [1].

Why flash floods need a local approach. In small, steep basins water can rise in less than an hour [5]. Storms are local and gauges sparse, so danger is often seen only when the water is already on the road, and road crossings are where people meet it: in the United States nearly half of flash-flood deaths are vehicle-related [6] (we found no Tunisian figure). The channels can also fail: bulk SMS can be delayed by hours in congested networks, and cell broadcast has no back channel, so nobody knows whether a message was received [7].

Why now. Tunisia's disaster-risk strategy to 2030 calls for people-centred early warning using *simple, low-cost equipment* (Action 8a, priority 1) [8]. National funding goes mainly to INM (US$24 million) [9], and the first pilot sites are Greater Tunis, Ghardimaou and Kebili, not Cap Bon [10]. The UN *Early Warnings for All* initiative aims to cover everyone by the end of 2027 [11].

1.2 Study site: Oued Souhil at the RR27 crossing

We use Oued Souhil, a wadi of about 7.4 km near Nabeul, as the illustrative site (Figure 1.1) [12]. It lies in the governorate hardest hit in 2018 and among the hardest hit in 2026 [3, 4]. It crosses the RR27 primary road on a bridge about 6.15 km along the channel, a concrete place where people meet the water [12]. Satellite imagery shows a sandy bed with industrial buildings beside the channel near the bridge [13], and an agricultural experimental station on the wadi is a possible partner for site access and data [14].

Three nodes are placed along the channel, at about 3.3, 4.9 and 6.15 km. Final node positions, and the choice of crossing itself, will be agreed with the municipality and Civil Protection through a field

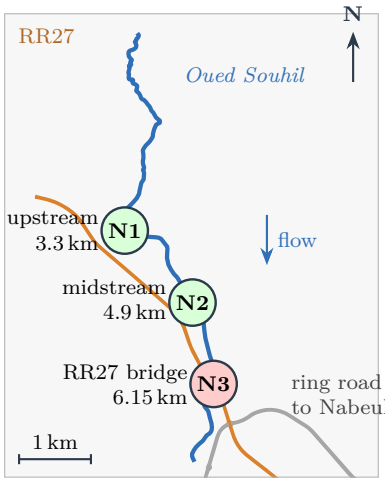


Figure 1.1: Study site: Oued Souhil and the RR27 crossing (plan view, north up), drawn from OpenStreetMap data [12]. Node positions are illustrative and given as distance along the mapped channel: N1 upstream sees the flood first, N2 confirms that the water is moving, and N3 at the RR27 bridge warns people where they would cross.

survey.

1.3 Existing solutions and their gaps

Detecting a flood is no longer the hard part: low-cost LoRa level nodes, SMS alerts and multi-level risk states are all published [15, 16, 17, 18]. What remains open is what happens *after* detection, in places with weak connectivity (Table 1.1).

Table 1.1: Existing solutions: what they do well and what they leave open for a flash flood at a Cap Bon road crossing.

Solution	What it does well	What it leaves open
National SMS alerts (GSMA "We Care") [19]	Reaches citizens through all three operators; alerts decided by the disaster committees.	One-way: no confirmation that anyone received or acted on the alert. No local sensing at a crossing.
National early-warning programme (INM) [1, 9, 10]	Upgrades forecasting and national observation (US$24 M).	National and forecast level; Cap Bon not among first pilots; alert dissemination named as a weakness [1].
Google Flood Hub [20]	Free AI forecasts; Tunisia is covered.	Flash-flood coverage limited to areas with recorded flood history; forecasts risk, does not deliver or confirm a warning.
Low-cost LoRa flood monitors [15, 16, 17, 18]	Level sensing, rate of rise, risk levels, SMS; one solar design works offline with a siren [18].	Alerts stop at "sent": no per-person confirmation, no escalation to municipality or Civil Protection, no link to the national chain. Mostly lab or simulation.
Commercial mass notification (e.g. Everbridge) [21]	Recipients confirm receipt; unanswered alerts escalate.	Subscription-based and internet-dependent, built for institutions, not for a wadi with no network.

The gap. In a search of literature and products between 22 September and 7 October 2026 we found no low-cost flash-flood system that combines a local alarm at the crossing needing no network; confirmation of receipt, even without mobile network; automatic escalation when nobody answers; agreement between independent signals before alarming people; and a bridge into the human-decided national alert channel. Chapter 2 details the comparison (Table 2.1).

1.4 Community need

People at a wadi crossing need a clear, local warning early enough to act, and the responsible authorities need to know that someone has acted on it. Detecting rising water, or sending a message,

is not enough: the system must check that the message reached a person, say what to do, and move the alert up the chain when nobody answers. This serves drivers and pedestrians at the crossing ("do not cross"), residents and workers near the wadi, volunteers and the municipal duty officer, and Civil Protection and the regional committee, who need trustworthy evidence to decide on a public alert.

1.5 Measurable impact

Phase 2 targets are measured on the prototype and match the verification plan of Chapter 2 (Table 2.5); pilot targets apply to a field deployment after the competition (Table 1.2). They measure the warning chain, not lives saved, which a short pilot cannot show.

Table 1.2: Impact metrics and targets.

Metric	What it shows	Target
Phase 2 prototype (bench and replayed floods)		
Detection to beacon	The crossing is warned at once, with no network	$\leq 5\,\mathrm{s}$ after the Danger level is reached
Radio delivery	Readings reach the gateway	$\geq 99\%$ with retries
Escalation on silence	An unanswered alert moves up the chain	Municipality reached within $3\,\mathrm{min} \pm 5\,\mathrm{s}$
Time to acknowledge	Alerts reach a person who responds	Median $< 1\,\mathrm{min}$ in drills
False alarms	One faulty signal cannot alarm people	No Warning from a single signal group
Silence is a fault	A dead sensor is never read as "safe"	Fault reported within 3 heartbeats
Offline operation	The chain survives internet and mobile outage	Beacon fires; mesh acknowledgement accepted
Local repair	Technicians can maintain it	Part swapped using the manual alone
Pilot after the competition (field)		
Reach	Households and sites in the agreed zone	$\geq 90\%$ reached directly or by a volunteer
Response in real drills	The chain works with real people	Median time to acknowledge $< 1\,\mathrm{min}$
Trust	False alarms are counted and published	Reported every season

1.6 Sustainable Development Goals

[**Project Name**] contributes to three SDG targets [22]: **11.5**, reduce deaths and people affected by disasters, including water-related disasters, the project's main goal; **13.1**, strengthen resilience to climate-related hazards such as the intense rainfall of 2018 and 2026; and **1.5**, reduce the exposure of the poor and vulnerable to extreme events, through volunteers and the assistance flag (Section 1.7). It also supports *Early Warnings for All* [11] and Action 8a of the national strategy [8].

1.7 Inclusion and local actors

Warnings that reach everyone. The same warning travels through several channels, so it reaches people whatever their age, ability or access to a phone: a flashing beacon (for people who cannot hear) and a siren (for people who cannot see) at the crossing, needing no network or phone; SMS in Arabic or French with a place and an action, readable on any basic phone; and volunteers, alerted by SMS or a radio handheld that works without the mobile network, who pass the instruction on in person to elderly residents and people without a phone. Volunteers from community groups often raise risk awareness better than officials [23]. A household can also ask, through an optional assistance flag, to be visited first. Joining is voluntary: the registry holds only a phone number, a

location, a language and the flag, with written consent, stored locally under Tunisia's data-protection law [24].

Local actors. The *Nabeul municipality* hosts the gateway, names the duty officer, and trains technicians to inspect nodes and swap parts with a written manual. *Civil Protection (ONPC)*, present through 24 regional directorates and 196 brigades [1], receives escalated alerts and validates message wording. The *regional committee under the governor* decides on public alerts under Law 91-39 [1]; it receives our evidence and keeps the final decision. *INM* and the *Oued Souhil experimental station* are possible data partners [10, 14], and volunteers and residents help choose node locations, test messages and take part in drills. Parts are locally replaceable, the software is open source and SMS goes only to a small registered group, so running costs are mostly maintenance time.

1.8 Requirements

The evidence above sets six requirements (Table 1.3); Chapter 2 traces every design choice and test back to them.

Table 1.3: Requirements and the evidence behind them.

ID	Requirement	Why (evidence)
R1	Work without mains power, internet or mobile network	Storms that cause flash floods also cut power and networks; bulk SMS can be delayed by congestion [7].
R2	Confirm receipt by a responsible person, and act when nobody answers	National SMS and cell broadcast have no back channel [19, 7]; alert dissemination is a named weakness [1].
R3	Give a safe action, not just a danger level	In 2018 a warning existed the day before, yet some residents stayed on rooftops [3].
R4	Limit false alarms	False alarms reduce trust in the next warning [25].
R5	Be low-cost and locally repairable	The national strategy calls for simple, low-cost equipment [8].
R6	Fit the institutional chain; never replace human decisions on public alerts	Public alerts are decided by the national and regional committees [1, 19].

Chapter 2

Technical Soundness

2.1 Design overview

[**Project Name**] is a warning *chain*, not a sensor. It detects a rising wadi, decides how dangerous the situation is, delivers the warning through whichever channel still works, requires a responsible person to confirm it, escalates automatically when nobody does, and hands the evidence to the authority that decides on public alerts. Every choice traces back to the six requirements of Chapter 1: work without grid, internet or mobile network (R1); confirm receipt and act on silence (R2); give safe-action guidance (R3); limit false alarms (R4); be low-cost and locally repairable (R5); fit the institutional chain and never replace human decisions on mass alerts (R6).

Four principles follow. **Local first:** the danger signal at the crossing is produced and shown at the crossing, and no remote layer is needed for the beacon to fire. **Silence is never "all clear":** a missing message raises a fault and never lowers the risk level. **Two independent signals before alarming people:** one reading can only raise a watch. **Humans own decisions:** the system recommends to the regional committee and never issues a public alert itself.

Figure 2.1 shows the architecture and Figure 2.2 how it sits on the ground. Three solar nodes are placed along one wadi: **N1** upstream, **N2** midstream and **N3** at a road crossing. On the illustrative site (Oued Souhil, Nabeul) they sit at about 3.3, 4.9 and 6.15 km along the mapped channel, the last at the RR27 bridge [12]. They report by radio to a gateway on a public building near the RR27 crossing (exact site to be agreed with the municipality), which hosts the network server, the risk engine, the escalation engine, the event log, the alert exporter and a read-only dashboard.

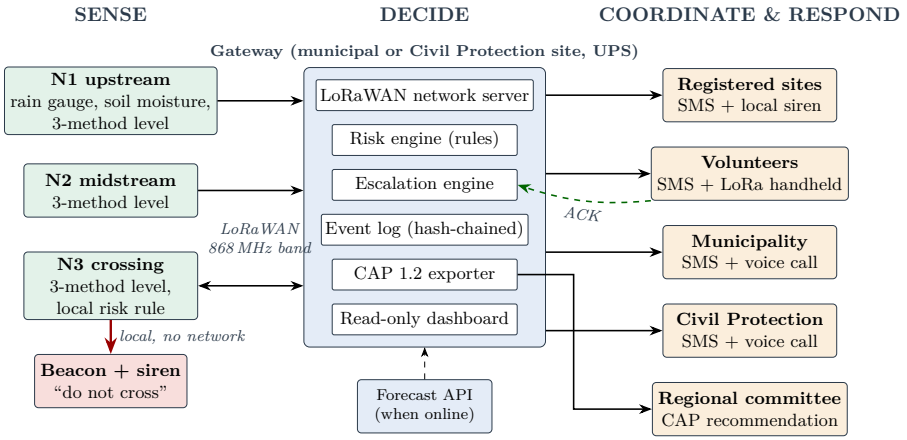


Figure 2.1: System architecture. N3 shows danger at the crossing even if the gateway is unreachable. The gateway adds the chain: rules across nodes, delivery, acknowledgement (dashed) and escalation, a tamper-evident log and a standard-format recommendation to the deciding authority.

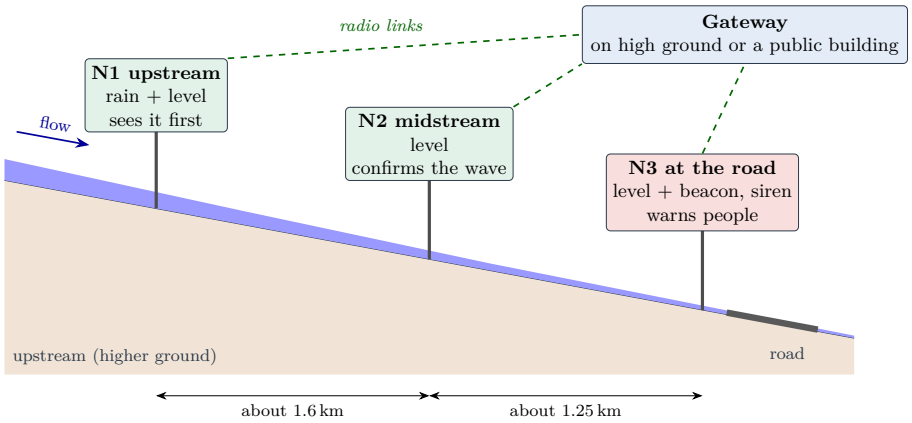


Figure 2.2: Side view of the wadi (not to scale). N1 sees the flood first, N2 confirms that the water is moving, and N3 warns people where they would meet it. Distances are from the illustrative site [12].

2.2 Innovation and efficiency

We do not claim a new sensor. Wadi level sensing over LoRa, SMS alerts, rate-of-rise monitoring and multi-level risk states are all published [15, 16, 17, 18], and acknowledgement with escalation exists in commercial, internet-based mass-notification platforms [21]. The closest system we found, a solar LoRa flood detector with online and offline modes, issues two severity tiers and was tested in the laboratory [18].

Our contribution is the **combination, designed for the conditions where warnings fail**: per-recipient acknowledgement with timed escalation from neighbourhood to Civil Protection; an acknowledgement path that needs no mobile network; agreement between independent signals before people are alarmed; fail-safe treatment of silence; and a Common Alerting Protocol (CAP) bridge into a human-decided national channel. In a search of literature and products between 22 September and 7 October 2026 we found no published low-cost flash-flood system combining these (Table 2.1).

Table 2.1: Comparison with the closest work found. ✓ = reported; – = not reported in the parts we read (absence of evidence, not evidence of absence).

Capability	[15]	[17]	[18]	[16]	[21]	**[Project Name]**
Level sensing over LoRa / LoRaWAN	✓	✓	✓	✓	–	✓
Rate of rise and multi-level risk states	–	✓	✓	–	–	✓
Local alarm without internet	–	–	✓	–	–	✓
Per-recipient acknowledgement, timed escalation	–	–	–	–	✓	✓
Acknowledgement without mobile network	–	–	–	–	–	✓
Independent-signal agreement before alarm	–	–	–	–	–	✓
CAP bridge to a human-decided channel	–	–	–	–	–	✓

Efficiency. Energy, airtime and money are spent only where they change outcomes: nodes sleep and sample faster only when risk rises; radio settings keep storm-mode reporting inside duty-cycle limits; SMS is reserved for a small registered group, since bulk SMS can be delayed [7] and public reach belongs to the national channel; and the rule-based logic runs on one single-board computer with no cloud dependency.

2.3 Sensing layer

Ultrasonic sensing alone can be disturbed by foam, turbulence and debris [26], and a node blinded during a storm is worse than no node. Each node therefore measures water level in three physically independent ways and votes (Table 2.2, Figure 2.3). The **ultrasonic sensor** hangs on an arm above the highest flood level and looks down: it times the echo of a sound pulse from the water surface and subtracts that distance from its mounting height. The **pressure sensor** sits at the bed inside a protective pipe: the deeper the water above it, the higher the pressure. The **float switches** are

fixed at three heights on a staff; each closes when the water reaches it, giving three yes-or-no answers that need no calibration.

Table 2.2: Sensing methods (manufacturer values, to be verified in our tests).

Method	Implementation	Weakness and mitigation
Ultrasonic level	Waterproof module on an arm (IP67, 3–450 cm, ≤8 mA [27])	Foam, spray, debris: median of 5 echoes, cross-check
Pressure level	Submersible 4–20 mA transmitter in a slotted steel pipe	Silt and impact: yearly cleaning
Float ladder	Three switches at Watch, Warning, Danger heights, supervised inputs	Three steps only: used as a hard check
Rain (N1)	Tipping bucket, counted in deep sleep	Clogging: mesh guard, monthly check
Soil moisture (N1, optional)	Capacitive probe, field-calibrated	Weakest signal: never triggers alone; removable

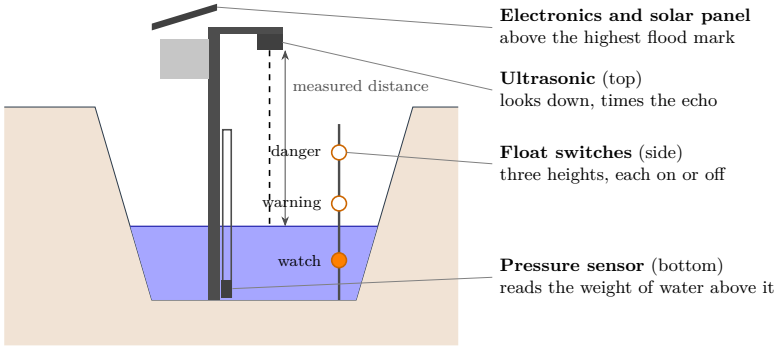


Figure 2.3: Cross-section of the channel at one node (not to scale): the three level-sensing methods are stacked at the same spot. Here the water has reached the Watch float only.

Fusion. The ultrasonic reading is the median of five echoes; the node cross-checks it with the pressure reading, and both are bounded by the float steps. If the two continuous sensors disagree for several samples by more than a tolerance that grows with depth (initially 30% of the depth or 10 cm, whichever is larger), the node raises a *degraded* flag and uses the *higher* of the two for alarms, never the lower. One rule decides Danger at a node: *two of the three methods above the danger height, counting only healthy methods.* If the other two are faulty or degraded, the Danger float alone is enough, so the node fails toward alarm. If the float is closed but both healthy continuous sensors agree that the water is well below it, the node raises Warning and a float-fault flag instead, so a float jammed by debris cannot alarm the town. A painted staff gauge at each node lets residents read the level themselves.

Soil moisture: an optional, deliberately weak signal. The probe at N1 is there for two reasons: the same rain produces more runoff on wet ground, so the system can react earlier; and in the dry season it records drought and water-resource data at almost no extra cost. It is also the weakest sensor in the design. One probe says little about a whole catchment, low-cost probes drift, and in semi-arid areas, including sites studied in Tunisia, runoff comes mainly from rain falling faster than the soil surface can absorb [28], so the rain gauge is likely the better predictor. We found no local data showing that it improves flood warning in Cap Bon. The design therefore gives it no authority: it can only adjust the rain thresholds, never raise an alert alone, and the system works unchanged without it. It is the first component to drop if cost or complexity must be reduced.

Electronics. Every node uses the same core, an ESP32-S3 microcontroller with a Semtech SX1262 LoRa transceiver (up to +22 dBm [29]), so firmware and spares are shared. A solar panel charges a lithium iron phosphate battery, chosen for its tolerance of heat inside a sunlit enclosure. Sensors are powered only while measuring; N3 adds a 12 V bus for the beacon and siren. Figure 2.4 shows the three interface circuits that matter most for reliability.

Firmware. At N1 and N2, sampling adapts to risk: every 10 minutes in Normal, every minute in Watch, every 10 seconds in Warning and Danger (initial values). N3, which protects the crossing,

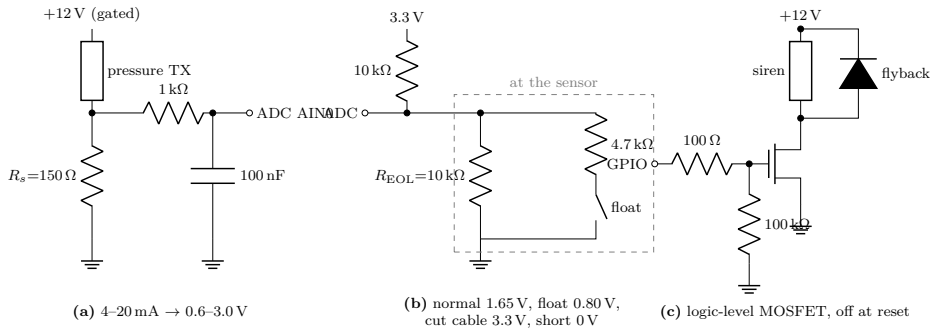


Figure 2.4: Interface circuits. (a) Pressure transmitter current read as a voltage by a 16-bit ADC. (b) Supervised float input: four distinct voltages, so a damaged cable reports a fault instead of "dry". (c) Beacon and siren driver, held off during reset.

never waits for the gateway to raise the risk level: it samples *every minute in all states* and every 10 seconds from Warning, so a storm that hits only the lower catchment, with no upstream signal and no gateway, is still seen within a minute. On every node a closing float also wakes the processor at once: the ESP32-S3 low-power coprocessor reads the supervised float inputs every second while the main core sleeps, and a high rate of rain-gauge tips wakes N1 the same way. N1 and N2 send a heartbeat at least every 10 minutes in Normal and with every report from Watch; N3 sends one every minute. The gateway marks a node *faulty* after three missed heartbeats, so a silent N3 is noticed within 3 minutes. N3 evaluates its own danger rule on every sample and drives the beacon and siren directly; it also listens continuously (LoRaWAN Class C), so the gateway can switch the beacon on early when N1 and N2 see the flood coming. Each node runs a hardware watchdog that resets hung firmware within seconds.

2.4 Communication layer

Network. Nodes use LoRaWAN, which gives per-device AES-128 keys, integrity codes and replay protection. A self-hosted ChirpStack server runs on the gateway, so the network works without internet [30]; a second gateway can be added for path diversity, as the server merges copies of the same message [31]. Each report is 16 bytes of data, 29 bytes on air.

Duty cycle. In the European rules commonly applied to this band, 868.0–868.6 MHz allows 25 mW at a 1% duty cycle and 869.4–869.65 MHz allows 500 mW at 10% [32]. Tunisian conditions are not yet confirmed: community tables conflict and the regulator's page is silent on this band [33, 34], so we are seeking confirmation and keep the band configurable. Table 2.3 shows the constraint: at SF10 a node could report only every 41 s. Storm-mode reports are therefore fixed at SF7–SF8 every 30 s, with state changes sent immediately as confirmed messages.

Table 2.3: Radio budget for a 29-byte frame at 125 kHz (airtime from the Semtech formula [35]; sensitivity from the noise-floor estimate with a 6 dB noise figure; margin over free-space loss on the longest node-to-gateway path, budgeted at 3 km, for +14 dBm, 2 and 5 dBi antennas, 2 dB cable loss).

	Airtime	Min. interval (1%)	Sensitivity	Max. path loss	Margin at 3 km
SF7	66.8 ms	6.7 s	−124.5 dBm	143.5 dB	43 dB
SF10	411.6 ms	41.2 s	−132.0 dBm	151.0 dB	50 dB
SF12	1646.6 ms	164.7 s	−137.0 dBm	156.0 dB	55 dB

Link feasibility. Nodes do not talk to each other; each one talks to the gateway, so the budget is set by the node-to-gateway paths, not by the spacing between nodes. With the gateway near the RR27 crossing, the straight-line distances on the mapped site are about 2.5 km for N1, 1.2 km for N2 and under 0.5 km for N3 [12]. We budget for 3 km to leave room for the exact gateway site: free-space loss is then about 101 dB, leaving 43–55 dB for terrain and vegetation. A comparable SX1262 network measured a 2.47 km link at −110 dBm [36], and LoRa path loss has been measured at 868 MHz in Tunisian oases [37]. Free-space margin is not enough on its own, because the nodes

sit on a valley floor and may have no line of sight to the gateway. Before installation, each path is therefore checked with a terrain profile (Radio Mobile on open elevation data) and then walk-tested. A path without line of sight gets a higher antenna mast at the node, a higher spreading factor (SF10 still allows one report a minute), or the optional second gateway on high ground upstream.

Graceful degradation. If the *internet* fails, everything local continues: rules on sensors only, SMS, voice calls, beacon and log. If the *mobile network* fails, the beacon and siren still fire, and volunteers still receive and acknowledge alerts on Meshtastic handhelds [38], an open-source LoRa mesh encrypted with AES-256 [39]; officials are alerted when the network returns, flagged as delayed. If the *gateway* fails, N3 notices that its confirmed messages are no longer acknowledged, keeps firing from its own rule, and shows a "gateway lost" status light for technicians; the maintenance team is told by the missing daily status message (Section 2.7). If *mains power* fails, nodes are solar and the gateway runs on its battery. If a *node goes silent*, the crossing is reported as "unmonitored", never as "safe".

2.5 Risk engine

The risk engine is a deterministic, rule-based state machine with four states (Figure 2.6). Signals form three *independent* groups: **M**, meteorological (a forecast flag from an open weather API [40] when online, or rain intensity at N1); **U**, upstream (level or rate of rise at N1 or N2); **D**, downstream (level or rate of rise at N3). One group can raise a Watch; a Warning needs two groups to agree; Danger needs two of the three healthy methods at N3 above the danger height (with the float fallback described under Fusion), or a danger level upstream with N3 rising. A restart never lowers the state (Section 2.7). The danger height is set from the site survey, relative to the road surface. All thresholds and hold times are initial values kept in one version-controlled file.

Why rules, and why more than one threshold. The engine uses no artificial intelligence: it compares measurements with thresholds. We chose rules over a trained model because there is no local training data and because Civil Protection must be able to read, audit and explain every decision. A single true-or-false test on one sensor would be simpler still, but it fails in four ways. One sensor can be wrong (foam, debris, a splash), so alarms need agreement. A level threshold at the crossing reacts only when the water is already there, so upstream rain and rate of rise are used to gain time. Water hovering near a threshold would switch the alarm on and off (Figure 2.5), so the alarm clears only below a lower level held for a set time. And a plain test reads a dead sensor as "below threshold", that is, safe; our rules treat silence as a fault.

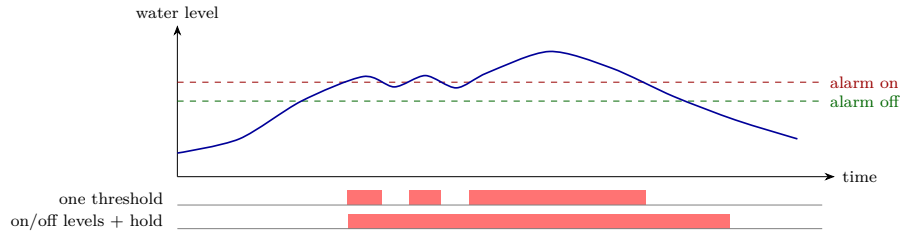


Figure 2.5: Why one threshold is not enough: the alarm (red bars) flickers three times while the water hovers around a single threshold; with separate on and off levels and a hold time, the same flood gives one continuous alarm.

Arrival estimate. The gateway records how long a rise takes to travel from N1 to N3 and, once calibrated on real events, can show an estimated arrival time as advice. We promise no lead time, which in small steep basins may be under an hour [5].

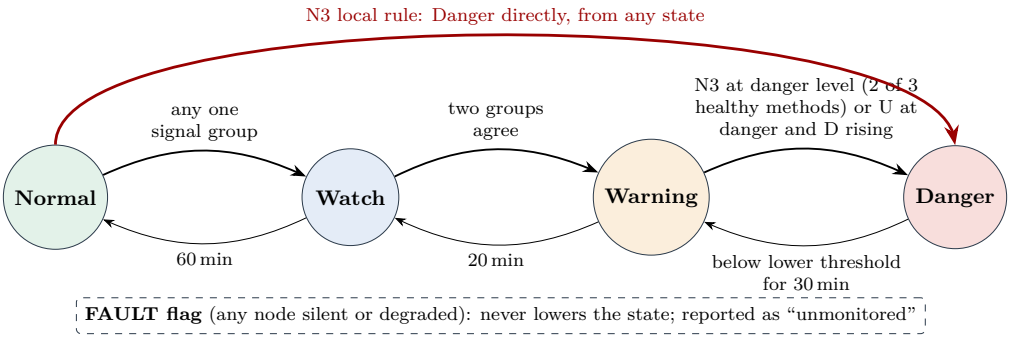


Figure 2.6: Risk state machine. Upgrades need more evidence as the state rises; downgrades need the signals to stay below a lower threshold for a hold time. Hold times are initial values.

2.6 Coordination: escalation and institution bridge

Each alert is a state machine with a timestamp on every transition (Table 2.4, Figure 2.7). Recipients answer `1` (received, acting) or `2` (cannot act); a `2` escalates at once. Volunteers answer by SMS or with the handheld button; officials are reached by SMS and, from tier T3, by an automated voice call answered with a key press. Only replies from registered numbers or paired handhelds count.

Table 2.4: Escalation ladder (timers are initial values to agree with Civil Protection).

Tier	Recipient	Triggered by	ACK	If no ACK
T0	Beacon and siren at N3	Danger (or N3 local rule)	–	–
T1	Registered sites	Danger	optional	–
T2	Zone volunteers (2–4)	Warning (information), Danger (action)	3 min	escalate to T3
T3	Municipal duty officer	T2 silent, or Danger lasting 10 min	3 min	escalate to T4
T4	Civil Protection, regional	T3 silent, or level still rising	3 min	repeat; log
T5	Regional committee	Danger: CAP recommendation, then updates	logged	–

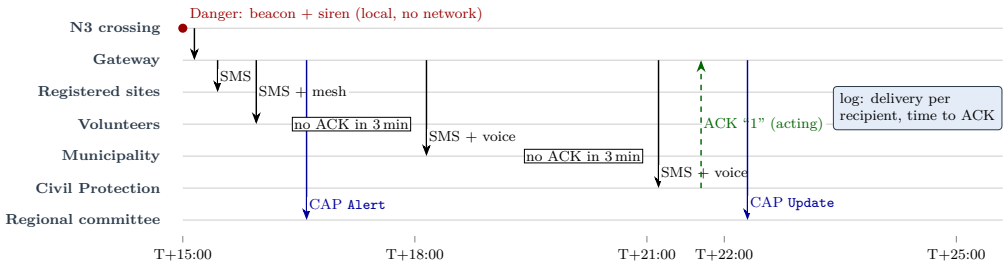


Figure 2.7: Escalation when nobody in the neighbourhood answers (illustrative times, not measured). The beacon fires first and needs no network; every message and acknowledgement is timestamped.

Message design. Messages carry a safe action, not just a level, because in 2018 a warning existed the day before and some residents still stayed on rooftops rather than cross floodwater [3]. Each message fits one SMS (160 Latin characters, or 70 in Arabic), names the place and says what to do, for example: *"DANGER Oued Souhil, RR27 bridge: do not cross. Move away from the wadi bed. Reply 1 if received."* Arabic and French texts are validated with Civil Protection before use.

Institution bridge. On Danger the gateway produces a Common Alerting Protocol 1.2 message [41] carrying the evidence (levels, rain, acknowledgement status, node health) and sends it to the regional committee, which decides on any public alert through the national channel [19, 1]. **[Project Name]** never sends public alerts itself. The message is signed, as CAP 1.2 permits [41], so the committee can verify its origin; demonstration messages use status `Exercise`.

Event log and dashboard. Every reading, state change, message and acknowledgement is appended to a local log in which each record contains the hash of the previous one, so later edits are detectable. The dashboard is a read-only view of this log; the system runs without it.

2.7 Power and resilience

The largest load is not the radio but waking the sensors every 10 s in storm mode. With our design assumptions (50 μA asleep, 100 mA for 1.5 s per sample, 120 mA while transmitting), N1 and N2 use about 9 mAh a day in Normal mode and 364 mAh in storm mode, so a 6 Ah battery gives about 13 storm days with no sun. The worst month at the site averages 4.5 kWh/m^2 per day on a tilted panel [42], so a 5 W panel returns about 4 Ah a day, roughly ten times the storm-mode need. N3 carries a 12 V, 20 Ah battery and a 30 W panel for the beacon and siren; sampling every minute and listening continuously add only about 0.2 Ah a day. Its processor has its own regulated supply with brown-out detection, so the current drawn by the siren cannot reset it. The gateway (about 10 W) runs on mains with a 100 Ah battery, about four days without power. All consumption figures will be replaced by measurements. Nodes are mounted above the highest recorded flood mark, in IP67 enclosures with hidden fixings and a tamper switch.

Restart during an event. A restart must neither lose an alert nor start it again from the beginning. The gateway writes every risk-state change and every alert (current tier, who was contacted, who acknowledged, next deadline) to its SQLite store *before* acting on it, and a service supervisor restarts any crashed service within seconds. At boot it restores this state: the risk level is the higher of the saved level and the newly computed one, and it can only go down through the normal hold times; each open alert resumes at the tier it had reached; timers that expired during the outage fire at once; recipients who had not yet answered receive one copy marked "resent after system restart", and those who had answered are not alerted again. The restart itself is logged and reported to the duty officer. At N3, the Danger state is also kept in memory that survives a reset. Outputs are off during reset (Figure 2.4c), but the firmware reads the sensors and the saved state within about a second of starting, and switches the beacon and siren back on if Danger still holds.

Health and maintenance alerts. Equipment problems go to a separate recipient, the municipal maintenance technician, so that they never mix with flood alerts. Each heartbeat carries the node's battery voltage and sensor status. The technician receives an SMS when a node's battery falls below 30%, a node goes silent or a sensor is degraded, mains power is lost at the gateway (with the estimated hours of battery left), and the gateway battery falls below 50% and 25%. At 25% the duty officer and Civil Protection are also told that the gateway will stop and that the crossing will then be protected only by N3's local rule. Before its battery runs out, the gateway saves its state, sends a last message and shuts down cleanly. A dead gateway cannot report its own failure, so it also sends the technician a short daily status message: if that message does not arrive, the gateway is down and is checked the same day.

2.8 Software, tools and methodology

Tools. Hardware is designed in KiCad and FreeCAD, with PVGIS for solar sizing. Firmware is built with PlatformIO and the RadioLib LoRaWAN stack. The gateway runs ChirpStack, an MQTT broker and our Python services: the risk and escalation engines as pure state machines, an SQLite event log, a CAP generator validated against the OASIS schema, and SMS and voice through the modem. The dashboard uses FastAPI and Leaflet with offline maps. Everything is open source and published on GitHub with continuous integration.

Methodology. We follow a light systems-engineering V-model: requirements R1–R6 are refined into system requirements, each with a design element and a verification test (Table 2.5). A failure-modes analysis of every component fed the degradation behaviour and the risk register (Chapter 4). Work runs in one-week sprints with a demo each week. The two state machines are tested before any hardware exists: property tests check, for example, that missing data can never lower the risk state, and a replay tool feeds recorded or synthetic flood curves into real nodes, so storms can be repeated on the bench.

2.9 Verification and feasibility

Table 2.5: Verification plan for Phase 2. Pass criteria are our targets.

#	Test	Pass criterion (target)	Req.
V1	Level accuracy against a ruler; foam and debris	Within 2 cm over 0–1 m; degraded flag, no false Danger	R4
V2	Terrain profile, then range walk test on each node-to-gateway path	≥99% delivery with retries	R1
V3	Power in each sampling profile	Within 20% of the budget	R1, R5
V4	Rain-only and splash scenarios (replay)	No Warning from a single signal group	R4
V5	Unanswered alert	T3 reached within 3 min ±5 s; every hop logged	R2
V6	Silent node, cut float cable, flat battery	Fault reported within 3 heartbeats	R1, R4
V7	Internet off, then mobile network off	Beacon fires; mesh ACK accepted; log intact	R1, R2
V8	CAP output; message comprehension; repair drill	Schema-valid and signed; safe action restated; part swapped with the manual alone	R3, R5, R6
V9	Storm at N3 only: N1 and N2 quiet, gateway off	Beacon on within 5 s of the Danger float closing	R1
V10	Danger rule: ultrasonic and pressure disabled with Danger float closed; then float forced closed with both sensors reading low	Danger in the first case; Warning and float fault in the second	R4
V11	Gateway power-cycled during escalation; N3 reset during Danger	Alert resumes at the same tier, nobody who answered is re-alerted, state not lowered; beacon back on within 2 s	R1, R2
V12	Mains off, node battery low, gateway switched off	Technician alerted within 5 min; missing daily status detected	R1, R5

Feasibility. Every component is off the shelf, and every software layer except our two state machines is a maintained open-source project. The hardware is deliberately simple (one core board for all nodes, common sensors, standard interfaces), which suits a team whose strength is software. The main risks are known and bounded: the radio band's legal status (configurable band, regulator consulted), sensor fouling (three methods, field tests), short lead time in small basins (local beacon, no lead-time promise) and coverage (terrain prediction, walk tests, second gateway). The Phase 2 prototype reproduces the full chain on a tray "wadi": three nodes and the gateway, a real radio link, the beacon and siren, real SMS and mesh acknowledgements, an unanswered alert escalating, a signed CAP file, and then the offline test V7.

2.10 Ethics in the technology

The commitments of Chapter 3 are built into the design. **Human authority:** the system recommends and the committee decides [19]. **Fail-safe:** silence raises a fault, outputs are off at reset, and a failed sensor reports "unmonitored", not "safe". **False alarms:** two independent signal groups and hold times protect trust, and false alarms are counted and published, because each one weakens the next warning [25]. **Privacy by design:** the household registry holds a phone number, a site, a language and an optional assistance flag, stored only on the gateway with consent under Tunisia's data-protection law [24]; the demonstration uses synthetic numbers. **Security:** encrypted and authenticated radio links, signed firmware and alerts, an allow-list for acknowledgements and a tamper-evident log. **Explainability:** every alert carries the readings that caused it, and the rules are public.

Chapter 3

Ethics and Values

A warning system decides who is told, who is asked to act, and what is recorded about them. Those choices are ethical before they are technical. This chapter states the commitments [**Project Name**] makes; Section 2.10 shows where each one is built into the design, and Table 3.1 lists the ethical risks we watch and how we check them.

3.1 Equity

Protection does not depend on registration. The beacon and siren at the crossing warn everyone present, including drivers passing through, seasonal workers and visitors who are on no list. Registration only adds a personal message; it is never the condition for being warned.

Free for the people protected. Households, volunteers and passers-by never pay for an alert, a call or a handheld. The system is paid for by institutions (Chapter 4), and the price to a municipality is published, so a poorer municipality knows exactly what it would need to fund.

No smartphone, no literacy barrier. Every alert exists as light, sound, a short SMS in Arabic or French, a voice call, and a person who comes to the door (Section 1.7). The assistance flag lets people with reduced mobility, elderly residents living alone and families with infants be visited first.

Sites chosen by exposure, not by influence. Where a sensor goes is decided by exposure: crossings with a record of flooding, traffic and nearby homes and workplaces, agreed with Civil Protection. A neighbourhood is not favoured because it is central, wealthy or able to pay.

Women and men take part equally. Message co-design sessions and drills are held at times and places that women, including women who work in the nearby industrial zone, can attend, and we aim for a balanced group of volunteers.

Table 3.1: Ethical risks and how we check them.

Risk	Safeguard	How we check
"No alarm means safe" (false sense of security)	Drills and messages say the beacon covers the crossing only; a failed node shows "unmonitored", never "safe"	Comprehension test in drills (V8)
People who are not registered are left out	The beacon and siren protect everyone; volunteers relay in person	Reach target $\geq 90\%$ (Table 1.2)
Trust lost after false alarms [25]	Two independent signals; false alarms published	Season report
Misuse of personal data	Opt-in, minimal data, stored on the gateway only, deleted 30 days after opt-out, declared under Law 2004-63 [24]	Yearly data audit by the municipality
The system replaces human judgement	The system recommends and the committee decides; the log records both	Decision log review
Burden on volunteers	Relaying only, right to decline, automatic escalation	Debrief after each event

3.2 Transparency

- **Open rules.** The thresholds, the Danger rule and the escalation timers are published in plain language and in the code. Anyone can see why an alert fired.
- **Open source.** Firmware, gateway software, schematics and the repair manual are published on GitHub under open licences.
- **Every alert explains itself.** Each alert carries the readings that caused it. Every recommendation sent to the regional committee is logged next to the decision the committee made, so the record shows who decided what.
- **A public season report.** After each flood season, the municipality publishes a short anonymised report. It covers alerts sent, false alarms, missed or late events, acknowledgement times, equipment faults and what was changed. Failures are reported as openly as successes.
- **Honest limits.** We do not promise lead time in small basins, and we say that the soil probe is a weak signal. The beacon protects the crossing; it does not protect the whole valley. People are told this in drills.
- **Open accounts.** The municipality and any funder receive the full cost of hardware, installation and service, and the sources of funding.

3.3 Sustainability

Environmental. The nodes run on solar power with lithium iron phosphate (LiFePO4) batteries, a chemistry without cobalt that lasts many years and is safe at high temperatures. All nodes use one core board, so spare parts are shared. Enclosures and mounts are reused when a node is upgraded. At end of life the batteries and electronics are returned to the team or the municipality and handed to an authorised e-waste collector, never left in the wadi. Mounting above the highest flood mark keeps the electronics out of the water.

Economic. The yearly running cost of a site is small and predictable: SMS and data credits, spare parts and a few hours of a technician's time (Section 4.5). It can be planned as a line in the municipal budget, so the system does not stop working when a grant ends.

Institutional. The system relies on open standards (LoRaWAN, CAP) and open-source software, so the municipality is not locked into one supplier, and any engineer can take it over. Technicians are trained with a written manual and must pass a repair drill (test V8), so maintenance does not depend on the original team.

3.4 Wellbeing

Lives first, and calm, specific messages. Every message names a place and an action ("Do not cross the RR27 bridge; use the alternative route") instead of a vague red warning. Specific instructions lower anxiety and make the right action easier. Requiring two independent signals before an alert keeps false alarms rare, so people are not woken for nothing.

No surveillance. The system measures water and rain, not people. It has no cameras and records no locations of people or movements. The only personal data is the opt-in registry described in Section 1.7.

Protecting volunteers. Volunteers relay warnings; they are never asked to go into water, close a road or stay near the crossing during a flood. They can decline an alert at any time, and the alert then moves on, because escalation is designed so that no single person carries the responsibility. Drills include a short debrief after each real event.

3.5 Community ownership

The municipality owns the system. The hardware, the data and the decision to operate it belong to the municipality, not to the project team. Our role after installation is support, through a service contract the municipality can end.

Residents shape it. Residents and workers help choose node locations, write and test the wording of messages, set the reply codes and take part in drills. A short community meeting after each flood season reviews the season report and decides on changes.

Local hands keep it running. Municipal technicians maintain the nodes, and volunteers check that sensors are clear of debris after storms. Both are trained and recognised publicly by the municipality. Knowledge stays in the community through the manual, the drills and the open repository.

Chapter 4

Management and Business

4.1 Mission, vision and values

Mission. Close the last kilometre of flash-flood warning in Tunisia: make sure that the people at a wadi crossing are warned in time to act, that a responsible person confirms every alert, and that nobody's silence is ever taken as safety.

Vision. By 2030, no one in Tunisia dies at a wadi crossing for lack of a warning that could have reached them.

Values.

- **Silence is never safe.** A missing reading, a dead node or an unanswered alert is treated as a problem, never as good news.
- **Humans decide.** The system gathers evidence and recommends; people with legal authority decide on public alerts.
- **Open and repairable.** Open designs, common parts and a written manual, so that local technicians can keep the system alive.
- **Everyone is warned.** Light, sound, SMS, voice and volunteers, so that no one needs a smartphone, literacy or registration to be protected.
- **Honest numbers.** We publish false alarms, faults and limits as openly as successes.

4.2 Team roles and organisation

The team has five roles. Each role owns a set of deliverables and backs up one other role, so that no task depends on a single person. The challenge requires members of IEEE SIGHT, the IEEE Systems Council and IEEE CEDA; the roles are matched to those memberships (Table 4.1).

How we work. We work in one-week sprints with a demo every Friday, a shared task board on GitHub Projects and a decision log. The PL chairs a 30-minute weekly review of schedule, budget and the top five risks. Technical changes go through pull requests reviewed by a second member. Any change to an alert rule must also be approved by the CP, who checks its effect on the people being warned. Table 4.2 shows who is responsible (R), accountable (A), consulted (C) and informed (I) for the main deliverables.

4.3 Implementation timeline

The plan has three stages: the Phase 2 prototype by the final on 10 December 2026; preparation of a field pilot in the first half of 2027; and installation in the dry summer, so that the system is tested before the autumn rains (the 2018 Nabeul flood came on 22 September [3]). Figure 4.1 shows the

Table 4.1: Team roles. Each role also acts as backup for the role in the last column.

Role	Owns	IEEE society	Backs up
Project and systems lead (PL)	Requirements R1–R6, V-model, integration, schedule, risk register, budget follow-up, submissions	Systems Council	CP
Embedded and hardware engineer (HW)	Node board, sensors, power and solar sizing, enclosures, node firmware, the N3 beacon and siren	CEDA	FV
Communications and software engineer (SW)	LoRaWAN and mesh network, gateway, risk and escalation engines, CAP, dashboard, GitHub and testing	–	HW
Field and verification lead (FV)	Site survey, terrain profiles and walk tests, calibration, tests V1–V12, repair manual, technician training	–	SW
Community, partnerships and business lead (CP)	Municipality and Civil Protection relations, consent and data protection, message co-design, drills, market, funding, video	SIGHT	PL

Table 4.2: Responsibility matrix for the main deliverables (R responsible, A accountable, C consulted, I informed).

Deliverable	PL	HW	SW	FV	CP
Phase 2 prototype (tray wadi, three nodes, gateway)	A	R	R	C	I
Verification tests V1–V12	A	C	C	R	I
Municipality agreement and data-protection declaration	C	I	I	C	A/R
Radio authorisation request (ANF)	A	C	R	C	I
Field installation and commissioning	A	R	C	R	C
Technician manual and training	C	C	C	A/R	C
Drills and message co-design	C	I	C	R	A
Budget, funding applications, season report	A/R	I	I	C	R

schedule and its milestones.

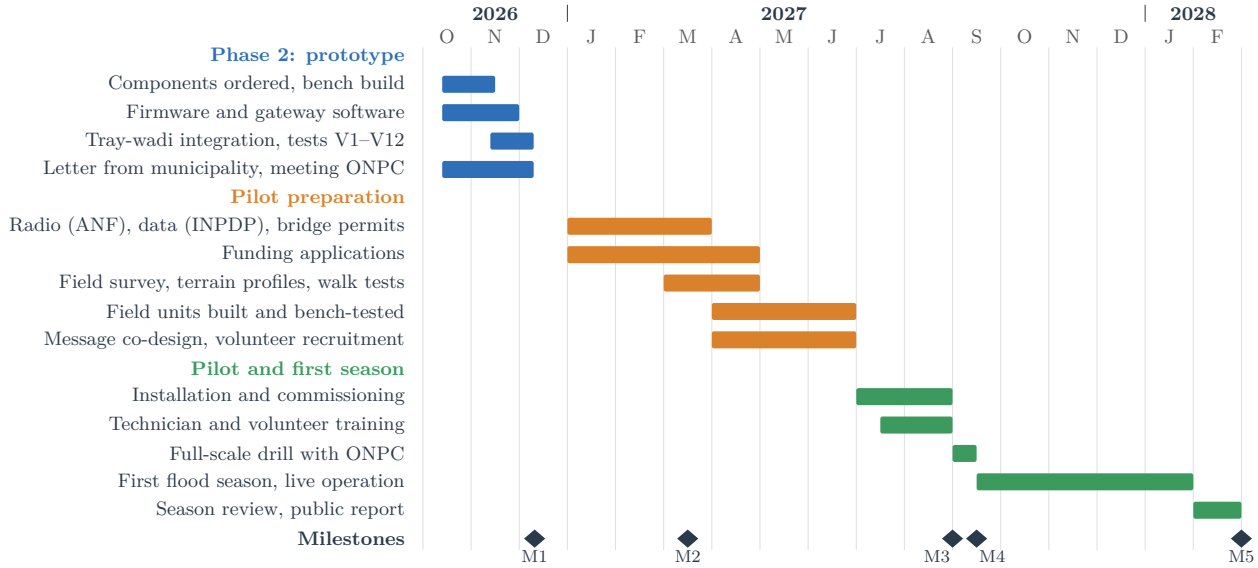


Figure 4.1: Implementation timeline, October 2026 to February 2028. Milestones: M1 final presentation and working prototype (10 Dec 2026); M2 agreement with the municipality and permits granted; M3 pilot installed; M4 first full-scale drill; M5 public report on the first season.

Why the timeline is realistic. The prototype uses only off-the-shelf parts and open-source software (Section 2.8). Components that must be imported are ordered in the first week after Phase 1, and every critical part has a second supplier. The permits are requested in January but are not needed until the July installation, which leaves three months of slack. The only fixed date is the start of the rainy season. If the pilot slips past it, we install N3 alone at the bridge first: it protects

the crossing without the rest of the chain, and the other nodes follow.

4.4 Market analysis

Demand. Floods cost Tunisia about US$40 million a year [2]. The World Bank names the dissemination of alerts as a weakness, and its Integrated Disaster Resilience Program finances disaster-risk reduction in the country [1]. The national strategy asks for people-centred early warning using simple, low-cost equipment (Action 8a) [8]. National early-warning funding goes mainly to forecasting at INM, and the first pilot regions do not include Cap Bon [10, 9]. The UN *Early Warnings for All* initiative aims for full coverage by 2027 [11]. Our product sits where these meet: a low-cost, last-kilometre system that turns a forecast or a rising wadi into confirmed action.

Customers and users. The one who pays, the one who operates and the one who benefits are often different (Table 4.3).

Table 4.3: Customer segments.

Segment	Role	What they gain
Municipalities	Owner, operator, buyer	Meets their duty to protect residents; a documented response chain; low running cost
Civil Protection (ONPC)	User, validator	Confirmed field evidence and escalated alerts; fewer rescues at crossings
Ministry of Equipment (roads)	Buyer for road crossings	Safer crossings on regional roads; evidence for closing a road
Industrial zones and large employers near wadis	Sponsor	Protection of workers who commute across the wadi; part of their duty of care
Development partners and DRR programmes	Funder of first deployments	A measurable, low-cost early-warning result aligned with Action 8a and Early Warnings for All

Competition and positioning. Table 4.4 compares **[Project Name]** with the alternatives a municipality could choose. National systems and commercial services are strong at what they do, but none of them warns the crossing itself without a network, confirms that a person acted, and escalates when no one answers. Professional hydrometric stations measure more accurately than we do, but they are designed for monitoring rivers, not for warning people, and their cost limits them to a few strategic sites. We do not compete with INM or the national SMS channel; we feed them evidence.

Table 4.4: Positioning against the alternatives (based on Chapter 1, Table 1.1).

	Sensing at the crossing	Local alarm, no network	Confirm and escalate	Link to national chain	Repaired locally	Low cost
National SMS system (GSMA) [19]	–	–	–	✓	–	✓
National EWS programme (INM) [9]	–	–	–	✓	–	–
Google Flood Hub [20]	–	–	–	–	–	✓
Commercial mass notification [21]	–	–	✓	–	–	–
Research LoRa flood monitors [15, 18]	✓	partly	–	–	✓	✓
[Project Name]	✓	✓	✓	✓	✓	✓

Market size. We size the market by sites, not by population. A site is one wadi crossing, or one exposed neighbourhood, with three nodes and a gateway. No public list of flood-exposed crossings

exists yet. In Phase 2 we will build one with Civil Protection and the Ministry of Equipment from their incident records. Until then we plan on conservative adoption: one pilot site in 2027, three new sites in 2028 and six in 2029, starting in Nabeul and the other Cap Bon municipalities, which were hit in both 2018 and 2026 [3, 4]. These are planning assumptions, not forecasts, and they drive the financial plan below.

Table 4.5: Strengths, weaknesses, opportunities and threats.

Strengths	**Weaknesses**
Works with no network; confirmation and escalation; open and locally repairable; very low cost per site	No field record yet; small team; depends on municipal commitment for operation
Opportunities	**Threats**
Action 8a of the national strategy; World Bank and UN programmes; Cap Bon not yet covered by national pilots; dry-season data for water management	Slow public procurement; radio authorisation; vandalism; loss of trust after a false alarm

Partners. The *Nabeul municipality* is the first host and owner. *Civil Protection (ONPC)* validates messages and receives escalations. The *regional committee* decides on public alerts. *INM* and the *Oued Souhil experimental station* are data partners [10, 14]. The *Ministry of Equipment* gives permission to mount N3 on the RR27 bridge. *ANF* rules on the radio band [34], and *INPDP* receives the data-protection declaration [24]. Local community associations recruit and support volunteers, and a university laboratory can host the test bench and student maintainers.

4.5 Budget and financial analysis

Assumptions. Prices are our estimates of Tunisian retail prices, or import plus shipping, at about 3 TND per US dollar. They are not supplier quotes yet; quotes for every item are a Phase 2 task, and a 15% contingency covers the difference.

Table 4.6: Hardware cost of one pilot site, in TND (estimates).

Item	N1	N2	N3	Gateway
Core board (ESP32-S3 with SX1262 radio)	75	75	75	–
Ultrasonic sensor, pressure transmitter, three floats, ADC	295	295	295	–
Rain gauge and soil probe (N1 only)	160	–	–	–
Battery (LiFePO4)	60	60	350	1 300
Solar panel and charge controller	80	80	200	–
Beacon, siren and driver	–	–	120	–
Enclosure, mast, pipe, staff gauge, cables	250	250	300	250
Antenna, cable, lightning arrester	60	60	60	150
Raspberry Pi, LoRaWAN concentrator with GPS	–	–	–	660
LTE modem (SMS and voice) and SIM	–	–	–	180
Mesh radios (gateway and three handhelds)	–	–	–	440
Subtotal	**980**	**820**	**1 400**	**2 980**

Table 4.7: Pilot investment and yearly running cost, in TND.

Investment (one site)		**Running cost (per year)**	
Hardware (Table 4.6)	6 180	SMS, voice and data credits	360
Installation (labour, concrete, transport)	1 200	Spare parts reserve (10% of hardware)	620
Training, drills, printed material	500	Technician time (about 6 days)	600
Contingency (15%)	1 180	Drill and training refresh	300
		Hosting, backups, domain	120
Total	**9 060**	**Total**	**2 000**

Phase 2 prototype. The tray-wadi demonstrator (three nodes with small batteries, a full gateway,

one pressure transmitter, the beacon and siren, two mesh handhelds, a PVC channel and a pump) costs about **2 700 TND**. Its gateway is reused in the pilot.

Cost of protection. Spread over a five-year life, a site costs about $9\,060/5 + 2\,000 \approx$ **3 800 TND a year** (about US$1 270), or about 10 TND a day to protect one crossing and the homes and workplaces around it.

Return on investment. Most of the benefit is lives saved and vehicles that never enter the water. Neither can be predicted honestly for one site, so we use a break-even argument rather than a forecast:

- The five-year cost of a site, about 19 100 TND (US$6 400), is **0.006%** of the US$100 million needed for recovery after the Nabeul flood of 2018 [1].
- It is less than the price of one new family car. The system pays for itself over its whole life if it keeps a single vehicle out of the water, or saves one rescue operation and the injuries that come with it.
- Globally, early warning is among the best-value adaptation investments: 24 hours' warning of a storm can cut damage by 30%, and US$800 million spent on early-warning systems in developing countries would avoid US$3–16 billion of losses a year [43]. Flash floods give minutes, not hours, so we claim the value of a life-safety system at the crossing, not property savings across the valley.

Business model. Alerts are always free for the people protected. Institutions pay for two things. A *turnkey deployment* is sold to the municipality or road authority at 12 000 TND per site (cost 9 060 TND); the price includes installation, training and the first drill. A yearly *service contract* costs 2 400 TND per site (cost to us about 1 400 TND; the municipal technician does first-line maintenance). It covers software updates, second-line repair, SMS credits, recalibration, the season report and a yearly drill. Because everything is open source, the municipality can end the contract and maintain the system itself; the contract must stay worth its price. An employer near a wadi can also sponsor a site, which the municipality then owns.

Table 4.8: Three-year financial plan, in TND (planning assumptions).

	2027	2028	2029
New sites (sites under service contract)	1 (0)	3 (1)	6 (4)
Grants (prototype and pilot)	11 760	–	–
Deployment sales (12 000 per site)	–	36 000	72 000
Service contracts (2 400 per site)	–	2 400	9 600
Revenue	**11 760**	**38 400**	**81 600**
Deployment cost (9 060 per site; prototype 2 700 in 2027)	11 760	27 180	54 360
Service cost (1 400 per site)	1 400	1 400	5 600
Fixed costs (part-time staff, travel, legal, certification)	3 000	9 000	15 000
Costs	**16 160**	**37 580**	**74 960**
Net result	**−4 400**	**820**	**6 640**
Cumulative	−4 400	−3 580	3 060

The plan reaches break-even during 2029 with ten sites (Table 4.8). The 2027 gap of about 4 400 TND is the pilot's first year of service, which we provide free. It is covered by competition prizes, university support or a sponsor. **Sensitivity:** if hardware prices turn out 30% higher than our estimates, the margin on a deployment almost disappears. In that case we raise the price to 13 500 TND, which is still low for a public buyer, or drop the optional soil probe and use a cheaper pressure transmitter. If only half the planned sites are sold, break-even moves to 2030, and the fixed costs are cut to match.

Funding the first sites. The pilot and the first deployments are funded from disaster-risk programmes rather than municipal budgets: the World Bank Integrated Disaster Resilience Program [1], funding linked to the national strategy [8, 9], and partners of *Early Warnings for All* [11]. After the first season, the season report gives a municipality the evidence it needs to put the running cost in its own budget.

4.6 Risk register

Risks come from the failure-modes analysis of Chapter 2 and from the partnership and funding steps above. Each risk has an owner (Section 4.2) and is reviewed weekly. Likelihood (L) and impact (I) are rated high, medium or low before mitigation; the colour shows the combined level (red high, amber medium, green low).

Table 4.9: Risk register.

Risk	L	I	Mitigation and fallback	Owner
Imported parts late (customs, stock)	H	M	Order in the first week; second supplier for every critical part; prototype can run with substitute sensors	HW
Municipality does not commit to operate	M	H	Early meetings, letter of interest, co-design and drills; start with N3 alone at the bridge under ONPC or road-authority hosting	CP
Radio band not authorised as planned	M	H	Ask ANF early [34]; band and power configurable in firmware; legal power and duty cycle by design	SW
False alarms reduce trust	M	H	Two independent signals; hold times; false alarms published and reviewed each season [25]	FV
Sensors fouled or buried by debris	H	M	Three sensing methods; fault detection; cleaning after each storm; anchoring in the bank	FV
Funding for the pilot not obtained	M	H	Several applications in parallel; reduced pilot with N3 and the gateway (about 5 000 TND)	PL
Flash flood faster than the warning chain	M	H	Local beacon fires within 5 s at N3; no lead-time promise in any message	HW
Theft or vandalism	M	M	Mount high, hidden fixings, tamper switch; community ownership; low-value parts	CP
No line of sight to the gateway	M	M	Terrain profiles and walk tests; taller mast; slower radio setting; second gateway	SW
Hardware prices higher than estimated	M	M	15% contingency; quotes in Phase 2; drop the optional soil probe	PL
Team members unavailable (exams, internships)	M	M	Backup role for each member; documentation in the repository; sprints planned around exams	PL
Data-protection non-compliance	L	H	Opt-in, minimal data, local storage, INPDP declaration [24]; synthetic data in the demo	CP

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